MediTwin

A digital twin of a hospital's medical supply chain that turns a supply shock into an approved, paid purchase order, inside one ASI:One conversation



4 agents · 1 conversation · verified on-chain

Fetch-A-Thon 3.0 · GCET Edition | ASI:One Agent Challenge: From Intent to Action

DevPair [Vijay Singh · Aryata Srivastava]

Shortages are found when the shelf is already empty

HERO SCENARIO

11.5 days

of insulin stock left

17 days

until the delayed supplier delivers

Synthetic data. Supplier SUP01 delayed 10 days, insulin demand +30%.

Discovered too late

By the time pharmacy notices, the lead time to fix it is longer than the stock that remains.

What-ifs live in spreadsheets

"If this supplier is late, what runs out, when, and what is the cheapest safe fix?" has no quick answer.

The fix is many manual steps

Purchase order, supplier email, transfers from other hospitals and the advance payment are all separate.

Target users: hospital pharmacy and procurement managers, store in-charges and supply-chain planners

One conversation, six steps from intent to action

1



Understand

ASI:One LLM turns plain English into a structured scenario

2



Sense

Inventory Agent serves the live digital-twin state

3



Simulate

1,000 Monte Carlo runs per plan, 21-day horizon

4



Decide

You pick a plan on an interactive card

5



Act

PO, supplier email and hospital transfers

6



Pay

Supplier advance in FET, verified on-chain

The LLM only understands intent. Every number comes from deterministic, tested code, and money moves only after explicit approval and independent verification.

Four specialised agents, all hosted on Agentverse



Agent Chat Protocol to ASI:One · typed uAgents messages between agents · watchdog with fallbacks

Supplier 10 days late, insulin demand +30%: the agent's answer

100%

simulated stockout if nothing is done

754

vials ordered from Himalaya MedSupply, plus 150 + 80 transferred

Rs 3.65 L

total plan cost (Rs 361,920 order + Rs 3,000 transfers)

0 / 1,000

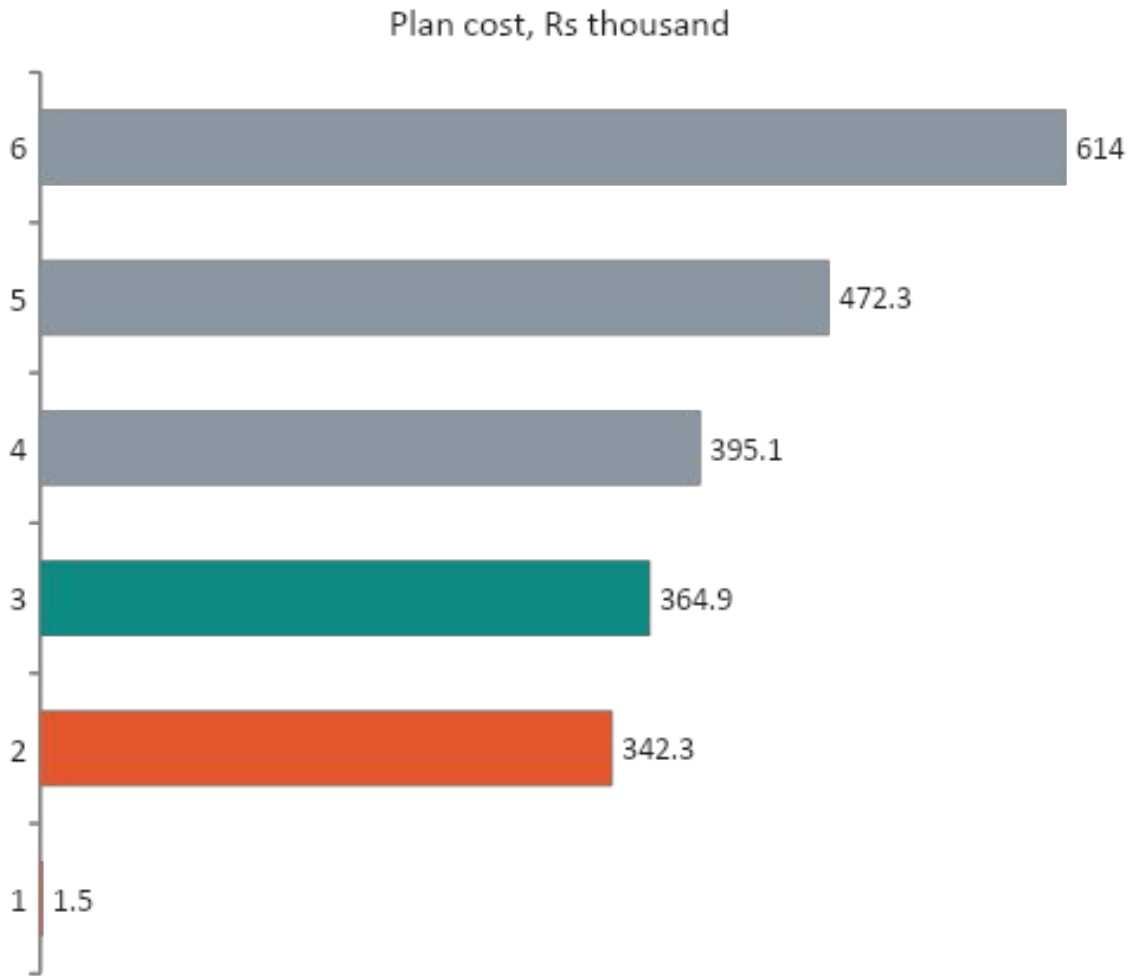
simulated stockouts after the plan (95% bound < 0.3%)



Cascade caught automatically: the same delayed supplier pushes Heparin to 9.8% simulated stockout risk.

Then: one click approves the plan, and the PO, supplier email, transfer requests and a payment card arrive in the same chat.

Cheapest is not safest: every plan is simulated first



Red: unsafe

Transfers alone (Rs 1,500) or a bridge through the delayed supplier still leave 98-100% simulated stockout.

Teal: recommended

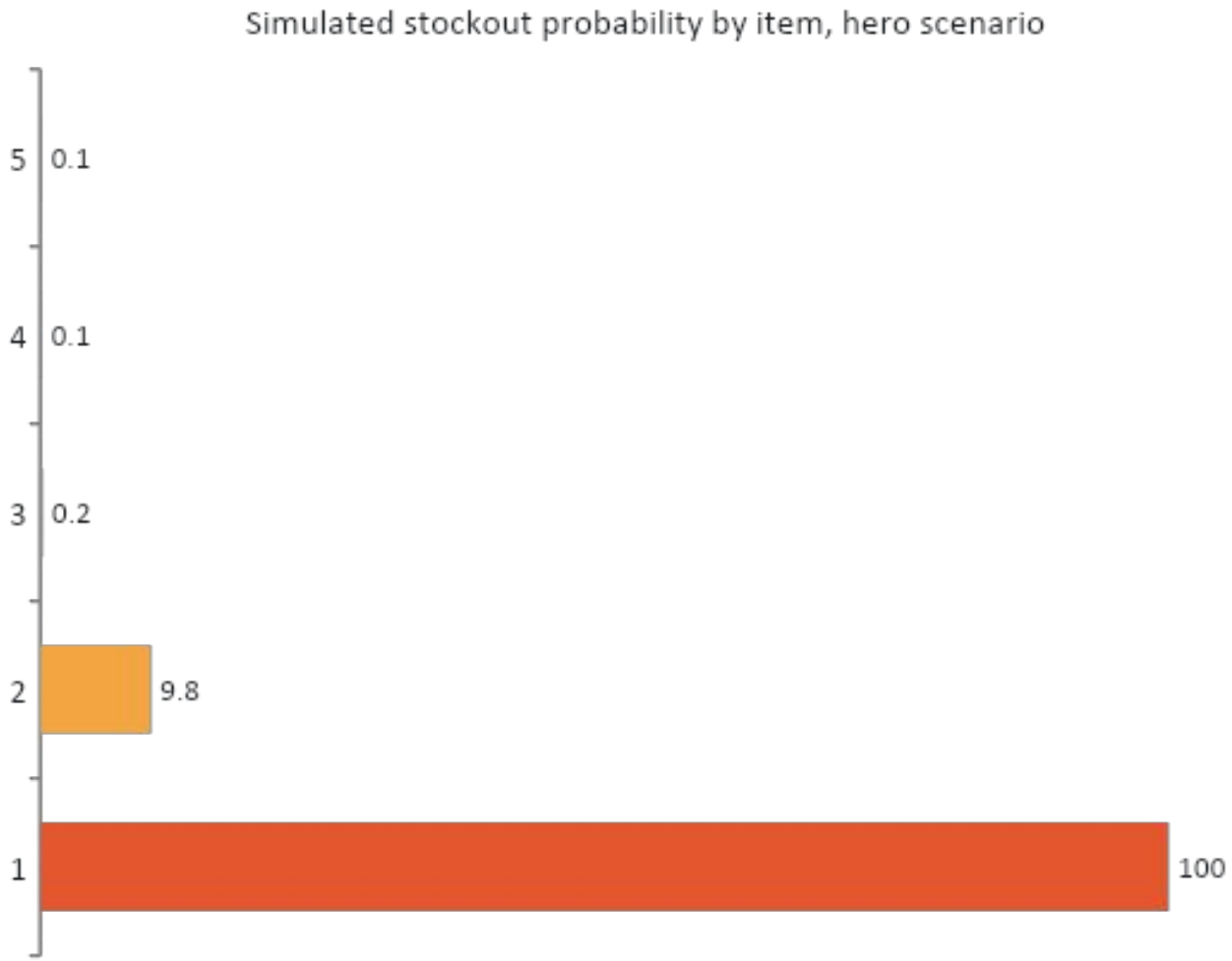
Cheapest plan with simulated risk at or below 5%: 0 of 1,000 runs stock out.

Grey: safe but costlier

Shown with cost and lead time so the human can choose.

If every safe plan exceeds the budget cap, the cheapest safe plan is still recommended and flagged for sign-off, never a cheaper plan that leaves the item at risk.

It also catches the cascade a spreadsheet would miss



One supplier, several items

Apex Pharma (SUP01) supplies both insulin and heparin. The twin simulates all 12 items under the same disruption, so the knock-on risk surfaces in the same answer.

12 items · 7 suppliers · 2 sister hospitals in the synthetic twin

A simulation engine built to be trusted

Common random numbers

One demand path and one delay draw per run, shared by every plan, so plan differences are real, not sampling noise.

Honest uncertainty

Run counts plus a one-sided 95% upper bound (Clopper-Pearson / Wilson). Zero stockouts is never shown as "0% risk".

Exact plan accounting

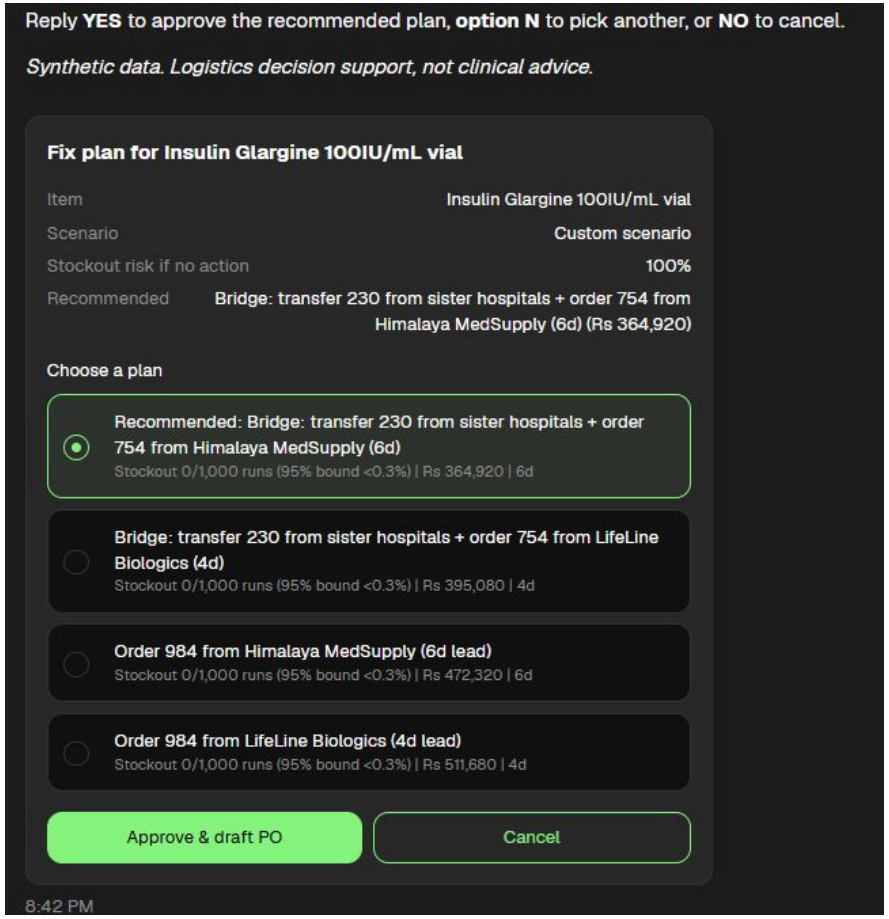
Supplier quantity, unit cost and cost kept separate from each transfer, and the same fields drive the PO and the payment.

Deterministic and tested

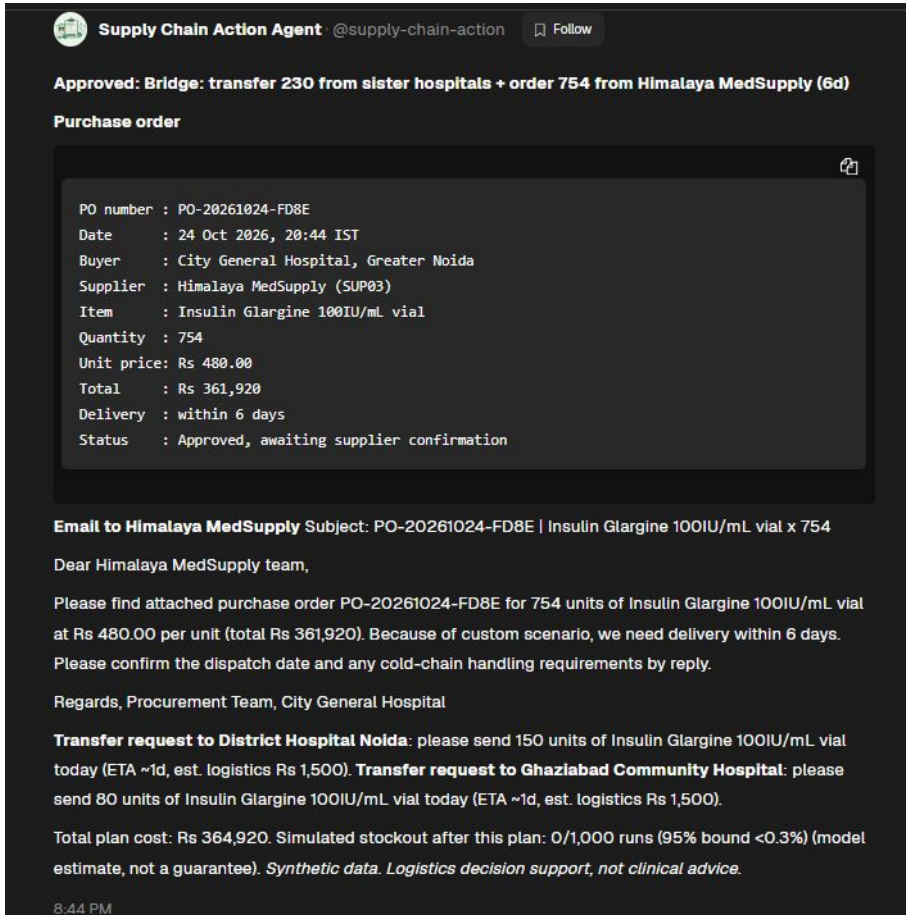
Seeded runs, 20 automated tests (hand-calculated edge cases included) and CI on every push.

Stated limits: no supplier capacity, minimum order quantities, expiry or delay distributions (a late delivery is a flat +3 days). Data is synthetic.

Decisions happen on Interactive Cards inside ASI:One



Plan picker with the recommended plan pre-selected



Approved PO: 754 x Rs 480 and named transfers

Card types

- Plan picker (detail card)
- Pay with FET or card
- Verify payment / cancel
- ASI:One native FET card

Clicks arrive as JSON or planner prose; a full text answer is always sent too.

Supplier advance paid in FET and verified on-chain

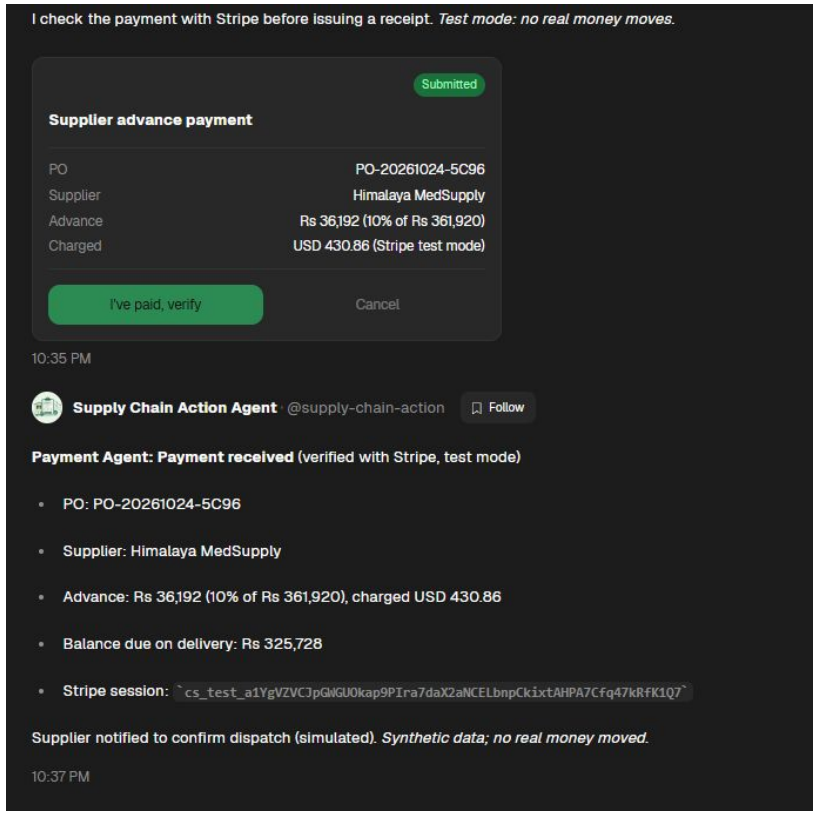
FET (default)

- Agent Payment Protocol, fet_direct
- Paid to the agent's own Fetch.ai wallet
- Verified on the Dorado testnet: success, recipient, amount
- Each transaction hash usable once

Card (fallback)

- Stripe Checkout in test mode
- Session paid, amount and PO must match
- Live keys refused; per-payment cap

Verified live, 25 Oct 2026: PO-20261025-BE8B paid in FET on the Dorado testnet (tx 26759D51…F612)



Shown: the card run (Stripe test mode). The FET run follows the same request, verify and receipt steps; 0.1 testnet FET is a demo amount.

Built end to end on the Fetch.ai stack



uAgents

Typed messages, interval tasks



Agentverse

4 hosted agents with profiles



Almanac

Agents found by address



Chat Protocol

ASI:One discovery and chat



ASI:One

Whole workflow in one chat



ASI:One LLM

Intent extraction engine



Interactive Cards

Plan, pay and verify cards



Payment Protocol

Seller role, full lifecycle



FET + agent wallet

fet_direct to the agent wallet



Ledger (cosmpy)

On-chain payment checks

Built to fail safely

Watchdog timeouts

Inventory silent 15 s: stored snapshot used. Simulation silent 40 s: clear message.

LLM circuit breaker

Quota or auth error pauses the LLM for 10 min; a strict keyword parser takes over.

De-duplication

Repeated deliveries and late replies are ignored by request and message ID.

Session-proof approvals

Approvals and payment clicks survive @mentions and ASI:One session changes.

Budget-safe advice

Never recommends an unsafe plan when a safe one exists; flags budget sign-off.

Replay-proof payments

Receipt only after ledger or Stripe verification; hashes single-use.

20

automated tests, run by CI on every push

+ offline payment checks with mocked Stripe and ledger

No keys or installs needed to run them

Why MediTwin fits each prize category



Best Multi-Agent Workflow

- 4 specialised agents with typed contracts
- Payment relayed agent-to-agent inside one chat
- Fallbacks when any agent goes silent



Best Personal AI

- Remembers preferred suppliers and budget
- Daily risk brief, tightest items first
- Every order and payment needs your approval



Best Agentverse Deployment

- All 4 agents hosted, in the Almanac
- Uses the hosted ASI:One LLM and agent wallet
- Tuned for hosted latency and storage

From hackathon prototype to a paid service

Monetization

- Per approved replenishment plan, or a monthly subscription per facility
- Collected through the same Payment Protocol flow, in FET
- Later: a small settlement fee on supplier advances

1. **Live ERP data**
   Replace synthetic data behind the Inventory Agent
2. **Disruption signals**
   Weather for cold chain, logistics delays, shortage notices
3. **Learned demand**
   Per-item demand from history
4. **Service fee in FET**
   Mainnet after compliance review
5. **Hospital network**
   Transfer optimisation across hospitals

From intent to action, in one conversation

Predict the stockout, choose the cheapest safe fix, approve it, and pay the supplier, all verified.

github.com/theajthakur/fetch-a-thon

Shared ASI:One demo chat

Demo video

Talk to it

@supply-chain-action

What if SUP01 is delayed 10 days and insulin demand rises 30%?

Agents: Action · Inventory · Simulation · Payment (@meditwin-payment)

Synthetic data · logistics decision support, not clinical advice · testnet payments